MTA

AUDIO EDITOR

Administrator Manual

Deployment, security, storage and operations



MTA Audio Editor 0.3.0-r210

Build: 20261015171900

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https://github.com/desalvo/mta-audio-editor

EUPL-1.2

Digital Audio Workstation with MTA support

Windows macOS Linux Android iOS

Operational guide for deployment, security, storage, backups, desktop/mobile clients, audio processing, MTA format, observability and releases.

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1. Architecture

MTA Audio Editor uses a FastAPI backend and a web frontend shared by browsers, desktop clients and mobile apps. Rendering, waveform processing, Demucs and MTA generation run on the backend.

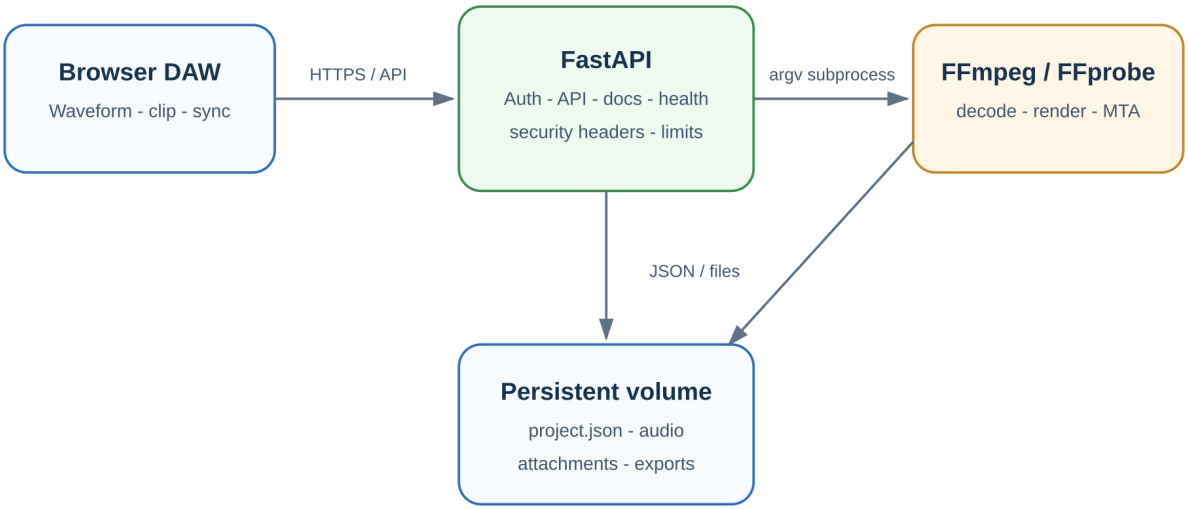


Figure 1 - Clients, API, processing and persistent storage.

Component	Platforms

Server	Linux, Docker, Kubernetes
Desktop	Windows, macOS Intel, macOS Apple Silicon
Mobile	Android, iOS/iPadOS
Browser	Modern browsers with Web Audio API

2. Deployment

Docker Compose should use a persistent volume for `/data/projects`. Kubernetes uses the supplied PVC and an optional Ingress. Application and Ingress upload limits must remain consistent.

```
docker compose up -d
# Kubernetes
kubectl apply -k k8s/overlays/nginx
```

TLS: production deployments should terminate HTTPS at a trusted reverse proxy or Ingress. Configure a stable public URL when email links or OAuth callbacks are enabled.

3. Configuration

Primary settings are supplied through environment variables and Secrets. Administrators should manage bootstrap credentials, upload size, SMTP, OAuth, session security and storage independently.

Area	Examples	Notes
Bootstrap	MTA_ADMIN_USERNAME, MTA_ADMIN_PASSWORD, MTA_ADMIN_EMAIL	Create the initial administrator
Upload	MTA_MAX_UPLOAD_MB	Keep proxy and application limits aligned
Public URL	MTA_PUBLIC_URL	Recommended behind proxies for stable callbacks/links
SMTP	host, port, TLS, credentials	Store credentials in Secrets

4. Storage and backups

The persistent root contains authentication/configuration data and project directories. Each project can contain its model, audio, attachments, originals, caches and analysis outputs. A full recovery backup must include the complete persistent root.

Recommended strategy: use consistent volume snapshots, keep multiple generations and test restores regularly. On Kubernetes verify that the StorageClass/CSI implementation supports the UID/GID and fsGroup used by the pod.

5. Security and users

Server deployments support local accounts, email verification, administrator approval, optional TOTP and optional OAuth providers. Social providers are disabled by default. OAuth requests are restricted to HTTPS and approved provider hosts.

Native single-user desktop mode bypasses the account database and login screens while retaining the internal authorization context required by local API paths. Secrets and signing material must never be baked into container images.

6. Audio pipeline

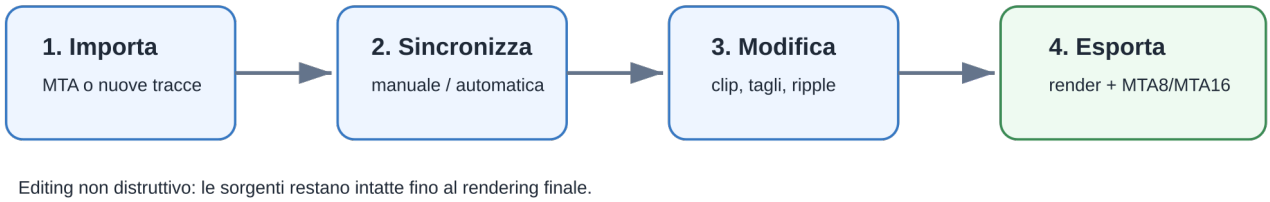


Figure 2 - Non-destructive editing and rendering flow.

FFmpeg and Demucs operations can be CPU/GPU intensive. Track previews are cached with signatures derived from source and relevant processing state. Long-running jobs expose progress and final artifacts through the API.

6.1 Dynamic preview and Render

Dynamic preview buffers track sources on the client and keeps them aligned to a reference clock. Render produces a server-side master. With RealTime meters enabled, track monitor branches feed analysers without becoming audible.

7. Desktop and mobile clients

Desktop clients include a local backend and native file pickers. Mobile apps connect to a remote backend.

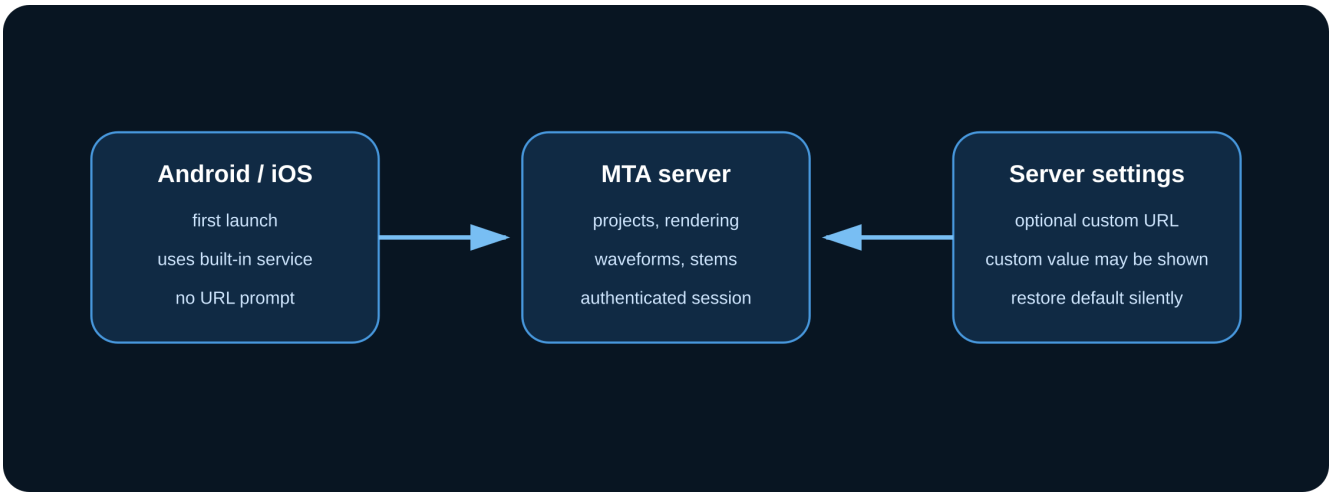


Figure 3 - Mobile connection and custom server settings.

Android and iOS automatically use the preconfigured service without prompting during installation or first launch. The built-in default address remains internal and is never displayed. A custom URL can be configured later and may be shown because it is explicitly user-selected.

8. MTA format

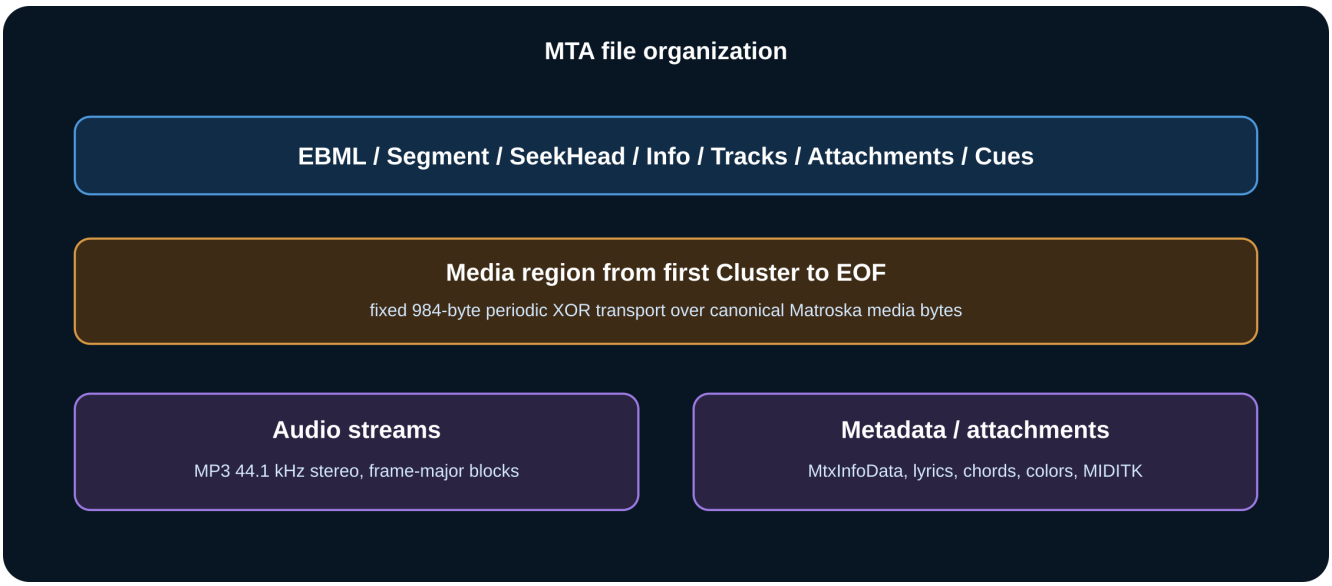


Figure 4 - High-level MTA container organization.

MTA uses EBML/Matroska structures, MP3 audio streams and metadata attachments. The media region starting at the first Cluster uses the documented periodic XOR transport. The complete bilingual specification is stored in `docs/MTA_FORMAT_FINAL_SPEC_IT.md` and `docs/MTA_FORMAT_FINAL_SPEC_EN.md`.

8.1 Click/Melody profiles

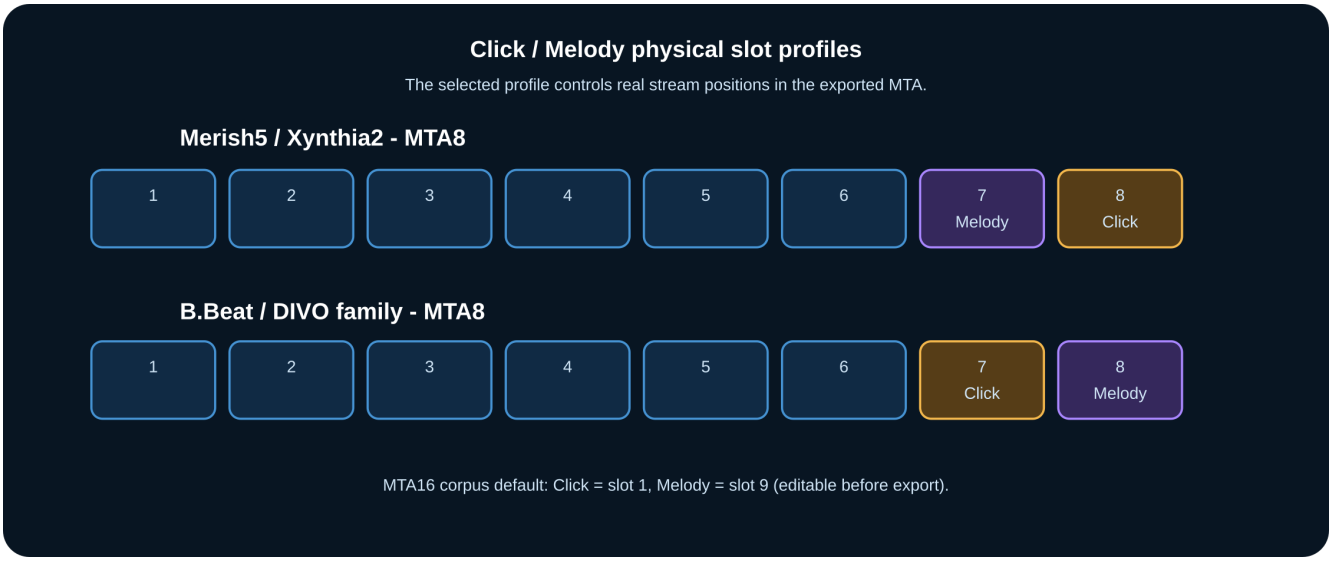


Figure 5 - MTA8/MTA16 export profiles.

MTA8 profiles preserve device-specific physical positions. The corpus-based MTA16 default uses Click 1 and Melody 9 and remains editable. Silent intermediate slots are generated when required to preserve the real stream number.

9. Operations and observability

Monitor health endpoints, persistent capacity, render duration, long jobs, application logs, FFmpeg/Demucs failures and service availability. Kubernetes probes should remain lightweight and independent from audio jobs.

Signal	Check
Health	`/api/health`
Storage	Free space and inodes
Jobs	Duration, state and server-side errors
Authentication	Login/OAuth/SMTP failures
Media	FFmpeg, Demucs, waveform/export logs

10. Upgrade and rollback

Before upgrading, create a backup, review release notes and production-gate results, update images/manifests and validate opening existing projects. For rollback, restore the previous application version and, when data compatibility requires it, the matching persistent-volume snapshot.

11. CI/CD and releases

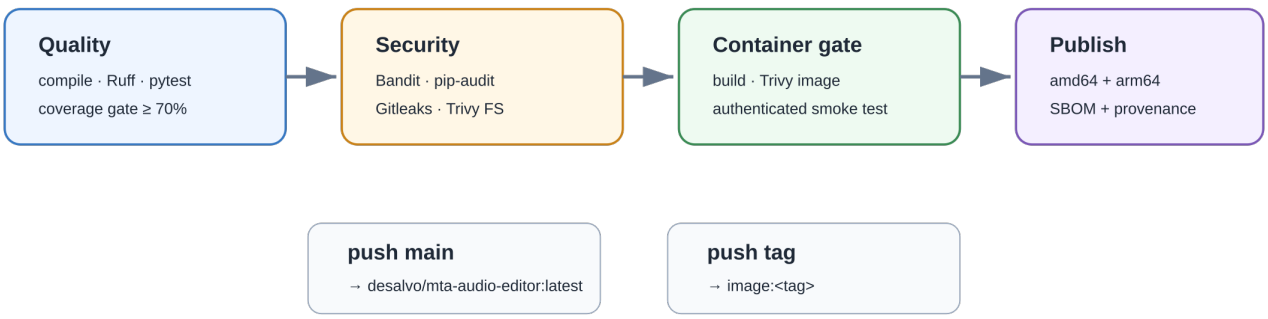


Figure 6 - Quality gates and packaging.

The production gate includes linting, tests, security checks and packaging validation. Native/mobile artifacts use dedicated build steps. Signing credentials belong in CI Secrets and must not be committed.

Release identity: `VERSION` stores the public product version (for example `0.2.0`), `REVISION` identifies the exact source/package revision (for example `154`), and `BUILD_INFO` remains the independent 14-digit build identifier. Stable tags use `v0.2.0`; artifacts may include `-r154`. Platform-required versionCode/CFBundleVersion values are technical packaging codes and do not replace `BUILD_INFO`.

12. Administrator troubleshooting

Symptom	Check
413 / upload rejected	Align application and proxy/Ingress upload limits
Stem job is very slow	CPU/GPU, RAM, temporary space and Demucs model
Login/email failure	SMTP, public URL, authentication logs and user status
Pod cannot write to PVC	Storage permissions, UID/GID and fsGroup
Mobile custom server unavailable	Custom URL, TLS, DNS and reachability from the device
MTA rejected by hardware	Slot profile, target capacity, firmware and target-device validation

13. On-device Demucs for iPhone/iPad - Core ML provisioning

The server can distribute Core ML-compatible Demucs models by setting `MTA_DEMUCS_COREML_MODEL_DIR` to a directory containing `demucs-2.mlmodel`, `demucs-4.mlmodel`, `demucs-6.mlmodel`, and/or `demucs-8.mlmodel`. The iOS/iPadOS client downloads models using the authenticated session, compiles them locally, and caches them in Application Support. Later separations can run offline on the device; Auto mode keeps the server as fallback for missing models, material over the local safety limit, or inference errors.

14. Native updates and Demucs models

Native Windows, macOS, Android, iPhone and iPad applications check GitHub for newer versions. The **Stable** channel uses tagged releases only, while **Early release** also includes the rolling `early-main` prerelease produced by the latest successful main build.

On iPhone/iPad the app automatically provisions a default 4-stem Demucs Core ML model: it prefers a bundled validated model and otherwise downloads it from the configured server. SHA-256 fingerprints are compared with the server inventory and installed models are refreshed when the fingerprint changes. Once installed, the local model remains available offline.

On iOS/iPadOS application updates must still use TestFlight/App Store or another operating-system-authorized distribution mechanism.

15. Automatic mobile Demucs model updates

The Docker/Kubernetes server periodically refreshes Core ML (iPhone/iPad) and ONNX (Android) model repositories from an HTTPS manifest, verifies SHA-256 and replaces files atomically. Mobile clients check the catalogue every six hours and at startup. **Update on Wi-Fi only** is enabled by default, so periodic model updates download only over Wi-Fi unless the switch is explicitly disabled. The baseline 4-stem model is provisioned at the first available network opportunity.

16. Extended stem model registry

Configure additional server models through `MTA_DEMUCS_MODEL_REGISTRY` or `MTA_DEMUCS_MODEL_REGISTRY_FILE`. Each profile declares `model`, `stem_count`, `stem_labels`, and optionally `engine`/`display_name`. Core ML/ONNX repositories are discovered dynamically from `demucs-N` files, up to 64 stems.

17. Dynamic Demucs model management

The server periodically refreshes the full configured model catalogue, including official Demucs models available in the runtime and Core ML/ONNX artifacts published by the manifest. Mobile and desktop clients download the selected model on demand when splitting is requested. Local options can delete, re-download, or force-update one or more models. Administrators can blacklist a model: managed server artifacts are deleted and the model disappears from the web/client catalogue until it is removed from the blacklist.

18. Runtime model tokens

`POST /api/models/token` issues revocable Bearer tokens limited to `models:read`. Core ML, ONNX and native bundle downloads require an authenticated session or such a token. CI uses assets from the GitHub `demucs-models` release rather than runtime credentials.

19. Lyrics/Chords engines and MP4

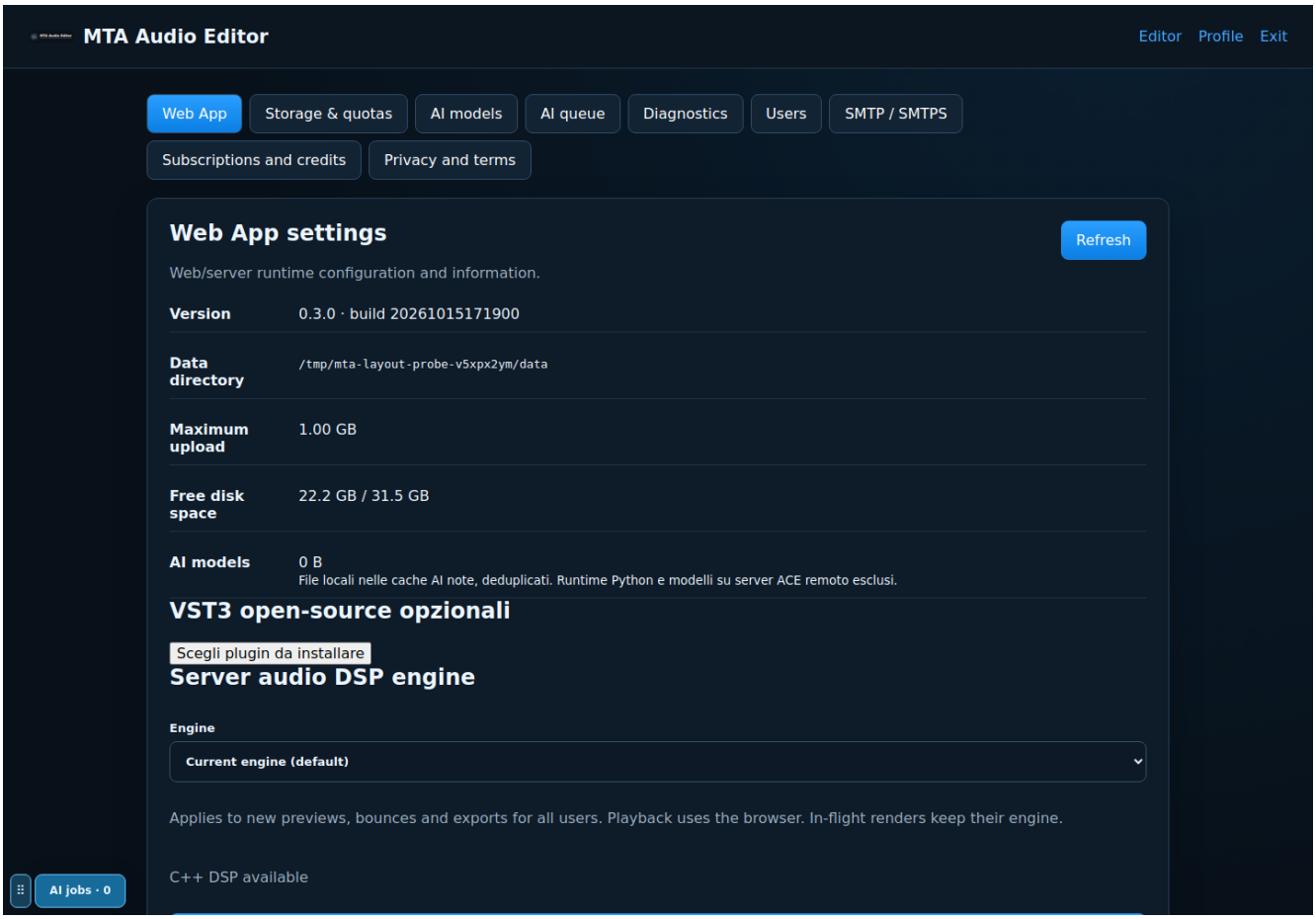
For maximum lyrics accuracy the default is `MTA_LYRICS_WHISPER_MODEL=large-v3`; `MTA_LYRICS_BEAM_SIZE` is also configurable; at temperature 0 a deterministic beam search with increased patience is used. For chords, Sonic Annotator with the Chordino/NNLS-Chroma plugin is preferred automatically when installed. MP4 export requires FFmpeg with the `ass`/libass filter and H.264/AAC codecs.

20. SIAE / Soundreef repertoire integration

SIAE and Soundreef are enabled by default in repertoire searches. Their official public pages always remain available as a fallback. For structured server-to-server lookup, configure authorized/licensed HTTPS endpoints through `MTA_RIGHTS_SIAE_SEARCH_URL` and/or `MTA_RIGHTS_SOUNDREEF_SEARCH_URL`. The endpoint must return JSON with a list under `results`, `items`, `works` or `data`; common fields such as title/original_title, authors/writers/composers, performers/artists, publishers/editors, ISWC, IPI and work_id are recognized. The application does not assume private or undocumented collecting-society endpoints.

21. r210 operational management

Settings groups Web App, users, diagnostics, SMTP, security, SSO and resource management. User search dynamically matches name, username or email. SMTP includes a test email: verify delivery before relying on notifications. Google, Facebook and GitHub remain disabled until configured; SSO identity does not replace admin approval. New registration requires email verification, second factor and approval according to the chosen enrollment flow.

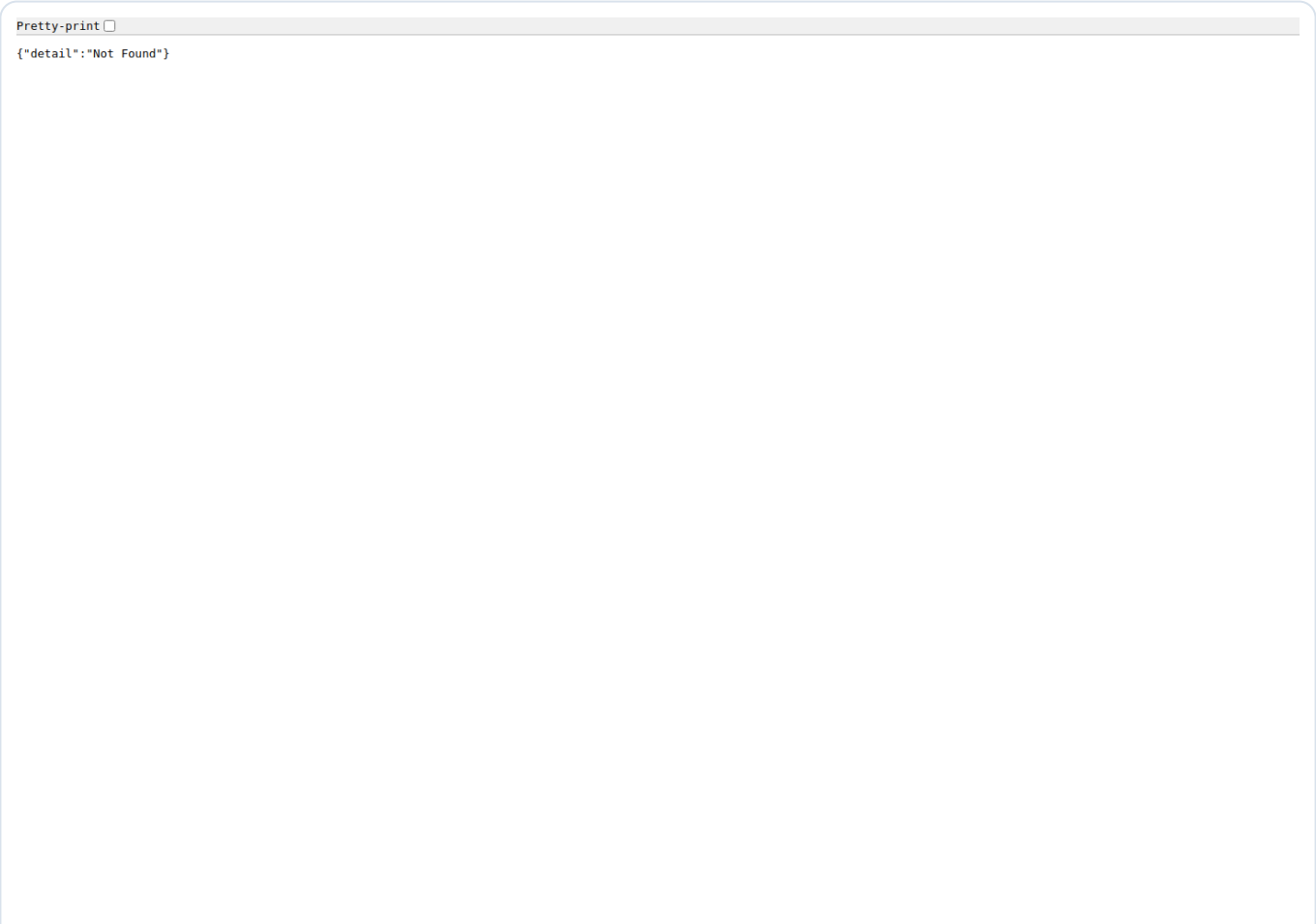


Administration interface on a disposable server without personal data.

Quotas, plans, credits and payments

CPU/GPU/storage quotas and rates are configurable. New users receive 5 GiB storage; admins have no default quota and cannot be disabled or deleted. A user ineligible for subscriptions retains accounting without billing or credits. Credits require the global option and an active/promo subscription; admins can grant promos and credits manually. Active-subscription limit changes retain the period and usage: increases require an explicit free/paid choice.

Termination can be immediate; the plan fee is prorated when billing is enabled. Storage extras follow paid additional allocated quota and its duration, not merely occupied bytes. Free extensions stay distinct. Configure rates/credit value and inspect period summaries. PayPal uses server-verified orders/captures; monetary refund adjustments are not automatically converted into credits.



Accounting and billing: check period, user and measurement origin.

AI runtimes and GPU

install_mta.sh installs AI engines into versioned /opt/mta/ai-* runtimes. These are needed at runtime: never manually delete the active environment. Automatic cleanup retains environments referenced by configuration. EL10 may provide FFmpeg through ffmpeg-free. SAM Audio/torchcodec/xFormers packages must match CPU/CUDA/ROCm; incompatible CUDA packages do not fix a ROCm environment. Diagnostics exposes installation status, measurement sources, artifact limits and error details.

In LXC, cgroup/GPU availability alone does not guarantee per-process counters. The host agent needs a mounted socket, permissions and kernel/driver compatibility. Zero or unverified counters must retain that status in reports. Before billing, verify measurement origin and the configured estimate policy. CI alone does not certify a production platform.

C++ DSP, VST3 and networking

C++ DSP and VST3 plugins are available on the server. RPM/DEB/Docker builds include the runtime; optional plugins are built/installed separately through the managed catalog. Licenses remain separate and need periodic review. A plugin used by a live session or open editor cannot be uninstalled. Original remote editors need the display backend; catalog presence does not imply all-plugin compatibility.

HAProxy/Ingress must allow streaming/WebSockets and avoid interrupting long jobs through proxy timeouts. The server runs DSP/AI while the browser plays received PCM: network buffering and DSP time are different metrics. Validate real transport after proxy, driver or plugin updates.

Retention, dumps and restore

Normal deletion retains accounting/billing and non-recyclable identities; project grace is configurable, default 60 days. Configured SMTP sends the archive link and expiry. Under deleted-user archives, admins can permanently remove remaining data with one confirmation. The privacy sandbox path has configurable retention (default 365 days) unless explicitly purged. Dumps can select users/projects; a user includes all their projects. Test restoration in isolation before production and retain an independent backup.

Privacy/terms are generated from instance data. Required fields are controller, country, support/complaints email, public HTTPS URL and effective date; missing optional fields are omitted. Legal-data changes notify registered users when SMTP is configured. Standard documents still require operator review for actual processing and jurisdiction.

22. MIDI/GM and AI studio deployment checkpoint

Editor/sidebar/MIDI-device preferences are isolated by identity and written lazily to the existing database. Independent patches merge transactionally. Historical native settings are not all migrated. A visible selector does not prove hardware compatibility.

GM requires a dynamic FluidSynth library and the reviewed FluidR3 GM 3.1 bank (148,398,306 bytes). Operator overrides: MTA_FLUIDSYNTH_LIBRARY and MTA_GM_SOUNDFONT_DIR. The bank checksum is checked before use. FluidSynth and the bank carry separate LGPL/MIT licenses: preserve their notices. Third-party binaries and bank data are not bundled in this source archive.

In Studio models, explicitly accept the linked Basic Pitch licenses and install its separate Python 3.11/3.12 ONNX CPU runtime. A real inference test and checksum verification precede activation. Administrators can test again, disable/enable or uninstall the model; active jobs prevent removal. Configure the job timeout (default 3600 s) and intermediate-result retention (default seven days). ONNX telemetry is disabled before initialization. Queue admission, accounting and project permissions are reused. No ROCm/CUDA compatibility or GPU consumption is claimed for Basic Pitch. Authorized singing models remain unavailable.

YourMT3 noPS is an additional operator-installed CPU backend for full-mix transcription into predicted GM/percussion parts. A real piano fixture passed through the queue and was saved as editable MIDI. This is not a ten-instrument accuracy benchmark. Velocity is generated; no calibrated confidence is reported. Installation, pinned hashes and licenses: docs/YOURMT3_RUNTIME_R210_EN.md. Vocal synthesis and multimodel reconciliation remain open.

Native Audio devices / ASIO / routing selects the host API, physical devices, buffer and stereo-master/input matrices. The compiled callback does not call Python. A duplex stream supports playback and stereo recording. Physical ASIO certification, independent per-track hardware buses and the ASIO corresponding-source redistribution bundle remain open. Native Studio models use dedicated local API paths; remote AI workflows use the shared token and recover results without duplicate charges.



Studio workflow and actual MIDI capabilities

Technical appendices

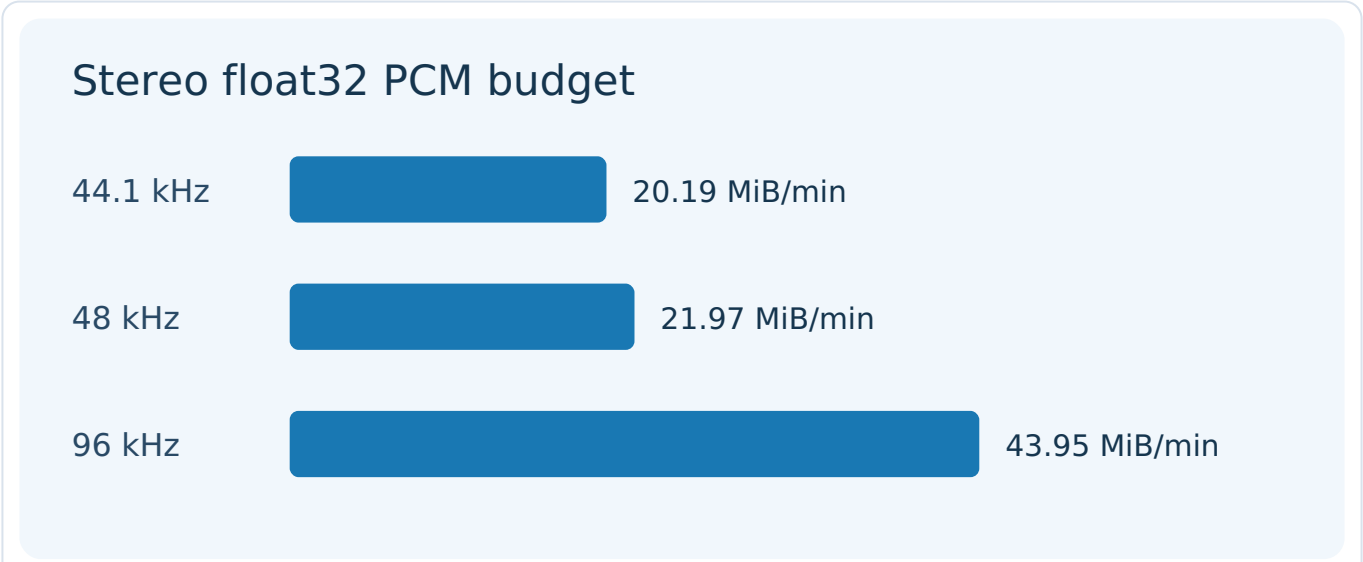
Documented state: 0.3.0-r210. Sizing guidance is an operational estimate, not a certified minimum. Local results do not certify physical devices, plugins or production servers.

A. Application types and capabilities

Type	Audio / DSP	AI	Plugin / MIDI
Web + server	Browser Web Audio; server C++ DSP	Global server queue; tokens for remote clients	Server Linux VST3, remote editor; Web MIDI where supported
Windows desktop	Current or C++; device transport	Local engines or token-authenticated remote server	Compatible Windows VST3; native MIDI
macOS desktop	Intel / Apple Silicon; Current or C++	Local models or server; accelerator support depends on engine	Matching-architecture macOS plugins; native MIDI
Linux desktop	Current or C++; local device	CPU / CUDA / ROCm depending on runtime and hardware	Linux VST3; ALSA/MIDI subject to availability
Android / iOS / iPadOS	Mobile shell; browser/ WebView audio	Server; on-device paths require a compatible model/ runtime	Server-side plugins; browser MIDI depends on platform

B. Resources and sizing

Profile	CPU / RAM	Disk / network	Note
Browser	4-8 GiB device RAM: planning baseline	HTTPS; stable network for streams and remote editors	256 MiB PCM cache; no guarantee for obsolete WebViews
Desktop / DSP	4 cores / 8 GiB starting point; 16+ GiB for dense sessions	SSD recommended; allow sources, exports and models	Plugins, sample rate and channel count determine actual load
Server editor	4 vCPU / 8 GiB for small tests; size for concurrency	Persistent volume, backups, bandwidth for concurrent sessions	Not a guaranteed user capacity
Server / desktop AI	16-32+ GiB RAM initial budget; model dependent	Separate weights/runtimes: several GiB, sometimes tens	VRAM/unified memory and precision depend on model; validate each engine
Mobile	Browser memory/CPU; avoid oversized projects	Stable Wi-Fi recommended for audio and remote plugins	Server DSP does not become phone-local



Theoretical PCM memory per minute, stereo float32 (excluding graphs, models and duplicate caches).

```
PCM bytes = seconds × sample rate × channels × 4
```

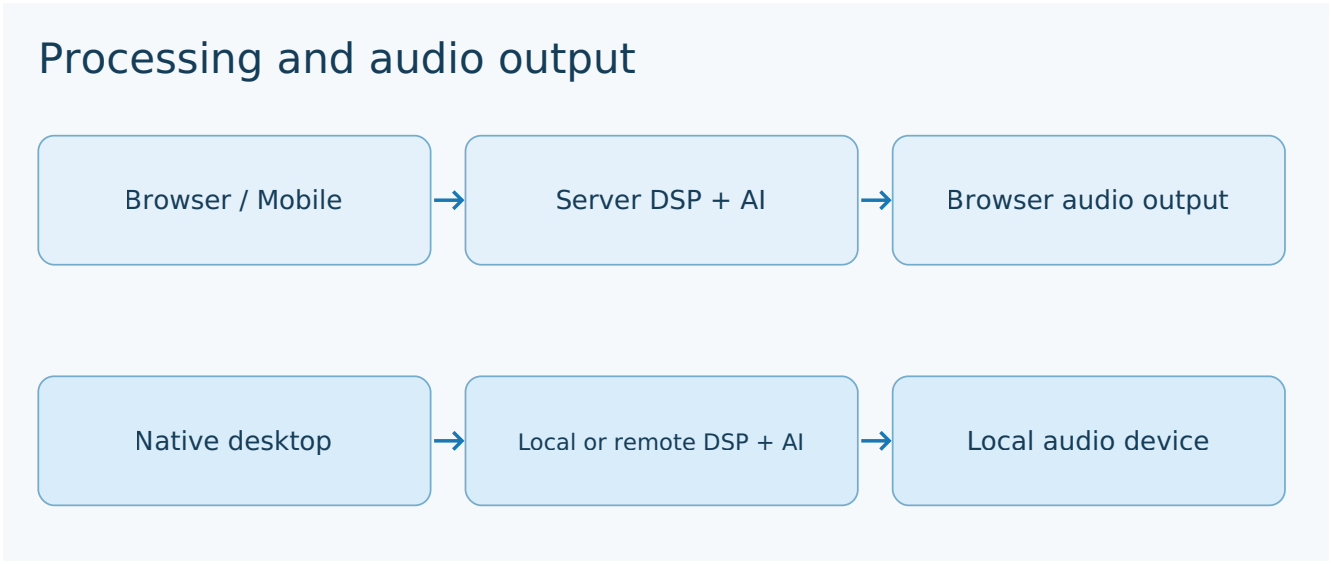
C. Limits and behavior

Area	Limit / behavior
Project	44.1 / 48 / 96 kHz; sample rate fixed after first track. Timeline up to 24 h.
Metronome	Standard, marker zones, adaptive. Pre-roll 0-64 bars; grid BPM 20-400 including decimals.
FX	Up to 16 inserts per chain; validate dependencies and VST3 compatibility per plugin.
MIDI	SMF0/1/2 up to 8 MiB; PPQ and SMPTE timebases. UMP/MIDI 2.0 depends on driver/provider, not guaranteed for every device.
AI	Global queue; configurable artifact limits and timeouts. An installed model does not guarantee GPU compatibility.
Streaming C++	Adaptive buffer 0.35-1.2 s plus block; not a guarantee of professional low-latency monitoring.
Storage / billing	New users default to 5 GiB; admins have no default quota. Quotas/credits depend on plan and permissions.

D. Observed performance and validation

Measurements from the preceding local checkpoint: Chromium, eight 30-second stereo tracks; test machine and load do not represent the production server. Original reports remain in the project validation directory.

Test	Observed result	Local source
C++ playback	Start 1.239 s; fader response 0.538 s; 0 underruns	r210-continuous-pcm-browser.json
C++ jitter / UI stalls	Network delays up to 180 ms, five 450 ms UI stalls: 0 underruns	r210-continuous-pcm-stress.json
Current engine	Start 2.198 s; cached restart 0.265 s	r210-continuity-current-engine.json



Processing and audio output locations.

Certification still depends on real-platform checks: audio interface, MIDI driver, third-party plugins, GPU and proxy. Accepted consumption is accounted; source and validity remain in the audit; zero consumption does not prove no GPU use.

GM instruments, singing and licences r210

From AI Studio, open Execution and Studio models to explicitly install the MIT FluidR3 GM bank and CC0 OddVoices voices, with status and progress. OddVoices is CPU concatenative singing, not a neural DiffSinger backend. Singers can render monophonic MIDI parts as vocalises or English syllables, saving audio and MIDI together. Italian/French and voice conversion are not declared available.

Accurate Audio-to-MIDI uses YourMT3 and Basic Pitch, retains original hypotheses and permits importing editable MIDI alternatives. It assigns no false note probability and does not guarantee accuracy for ten arbitrary instruments. Maximum accuracy remains disabled.

Native apps and local processing are free. Any server charges reimburse CPU/GPU/disk infrastructure costs without profit; model licences still apply. Policies, disabled components and manual activation conditions are documented in docs/MUSIC_ASSETS_AND_LICENSES_R210_EN.md in the package. The new verify_music_hardware.py script prepares ASIO/ROCm testing; inventory is not certification.



Installer and bank status after actual CPU tests, with collapsible licence policy review.

23. SAM Audio: service access and diagnostics

SAM Audio checkpoints are gated. With the same Hugging Face account, open SAM Audio Base and SAM Audio Large, accept the terms and wait for approval where required. The personal token must allow reading authorized gated repositories. A `GatedRepoError` HTTP 401/403 indicates authentication or authorization failure; it does not establish a GPU fault.

The server process must be able to use the login and cache; a login performed only under another shell user may not be visible. Inspect the effective settings without printing token values:

```
sudo python3 - <<'PYCODE'
import os, pwd, subprocess
pid = int(subprocess.check_output(
    ['systemctl', 'show', 'mta-server.service',
     '-p', 'MainPID', '--value'], text=True))
if pid <= 0:
    raise SystemExit('mta-server.service is not running')
with open(f'/proc/{pid}/environ', 'rb') as stream:
    entries = stream.read().split(b'\0')
env = dict(item.split(b'=', 1) for item in entries if b'=' in item)
print('User:', pwd.getpwuid(os.stat(f'/proc/{pid}').st_uid).pw_name)
for key in (b'HOME', b'XDG_CACHE_HOME', b'HF_HOME', b'HF_TOKEN_PATH'):
    print(key.decode(), ':', os.fsdecode(env.get(key, b'(not set)')))
for key in (b'HF_TOKEN', b'HUGGING_FACE_HUB_TOKEN'):
    print(key.decode(), ':', 'set' if env.get(key) else 'not set')
PYCODE
```

On the inspected server the service uses `root`, `HF_HOME=/var/lib/mta/models/huggingface`, no custom `HF_TOKEN_PATH` and no token in its environment. The SAM Python was `/opt/mta/ai-7f176167db5671ae/sam/bin/python`. That path is installation-specific: locate the current runtime before using it. With these values, authenticate and verify only the configuration files:

```
SAM_HF=/opt/mta/ai-7f176167db5671ae/sam/bin/hf
sudo env HF_HOME=/var/lib/mta/models/huggingface \
  "$SAM_HF" auth login
sudo env HF_HOME=/var/lib/mta/models/huggingface \
  "$SAM_HF" auth whoami
sudo env HF_HOME=/var/lib/mta/models/huggingface \
  "$SAM_HF" download facebook/sam-audio-base config.json
sudo env HF_HOME=/var/lib/mta/models/huggingface \
  "$SAM_HF" download facebook/sam-audio-large config.json
```

For a non-root service, run the commands as its actual user with the corresponding `HF_HOME` and optional `HF_TOKEN_PATH`. An environment `HF_TOKEN` overrides the saved token; also check for `HUGGING_FACE_HUB_TOKEN`, whose behavior depends on the installed version. These checks download only `config.json`; first inference can require the full weights and sufficient disk space.

> **Credentials:** never paste tokens into chat or tickets or print them in logs. Protect credentials for the service account. Checkpoint access and real GPU compatibility are separate checks.

If the error persists, inspect `sudo journalctl -u mta-server.service --no-pager` and share only necessary lines after redacting credentials and personal data. Structured diagnostic bundles can report the stage without the original exception.